Lumos MES - Information Architecture & Page Navigation Guide
1. System Overview
Lumos MES is an enterprise-grade Manufacturing Execution System designed for regulated industries (Food & Beverage, Life Sciences, and Pharmaceuticals). It enforces GAMP 5 validation principles, FDA 21 CFR Part 11 electronic records/signatures, GS1 supply chain serialization, and FSMA Section 204 Critical Tracking Events.
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2. High-Level Architectural Hierarchy
```mermaid
graph TD
    A["🔐 /login (Authentication & JWT Issuance)"] --> B["🖥️ Protected Layout Shell (Top Nav + Left Accordion Sidebar)"]

    subgraph Operations ["1. Operations Pillar"]
        B --> OP1["/dashboard (Executive Dashboard)"]
        B --> OP2["/monitoring (Production Monitoring & Status Legends)"]
        B --> OP3["/schedule (Gantt Scheduling & Timeline)"]
        B --> OP4["/operator (Operator Workstation & Touch Mode Console)"]
    end

    subgraph MaterialsRecipes ["2. Materials & Formulation Pillar"]
        B --> MR1["/material-management (Material Master, Vendors, GRN, FEFO Stock, Dispensing)"]
        B --> MR2["/recipes (Master Formulas, BOM Specifications & GAMP 5 CPP)"]
    end

    subgraph QualityCompliance ["3. Quality & Compliance Pillar"]
        B --> QC1["/work-orders (Manufacturing Batch Records & EBR Dossiers)"]
        B --> QC2["/quality (Quality & Statistical Process Control - SPC)"]
        B --> QC3["/equipment (Asset Register, Calibrations & IQ/OQ/PQ Qualifications)"]
        B --> QC4["/compliance (21 CFR Part 11 Electronic Signatures & Audit Logs)"]
    end

    subgraph TrackTrace ["4. Track & Trace Pillar (GS1 / FSMA 204)"]
        B --> TT1["/trace/dashboard (Traceability KPI Dashboard)"]
        B --> TT2["/trace/products (GTIN Product Registry)"]
        B --> TT3["/trace/batches (Batch Lot Campaigns)"]
        B --> TT4["/trace/trace (Multi-tier Genealogy Graph)"]
        B --> TT5["/trace/events (Critical Tracking Events CTE/KDE)"]
        B --> TT6["/trace/shipments (Distribution Ledger & SSCC-18)"]
        B --> TT7["/trace/recall (Targeted Mock Recall Engine)"]
    end

    subgraph Administration ["5. Administration Pillar (GAMP 5 Role Matrix)"]
        B --> AD1["/admin/material (Material Master Config)"]
        B --> AD2["/admin/equipment (Plant & Equipment Asset Hierarchy)"]
        B --> AD3["/admin/types-uom (Material Types & Standardized UOM)"]
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        B --> AD4["/admin/users (User Accounts & Role Mapping)"]
        B --> AD5["/admin/standards (GS1/Part 11/EPCIS Webhooks)"]
        B --> AD6["/admin/api-docs (REST API Playground)"]
    end

    subgraph Support ["6. Support Pillar"]
        B --> SP1["/help (Help Center & Standard Operating Procedures)"]
    end
```
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3. Complete Navigation Sitemap & Route Directory
Pillar / Group	Route Path	Page Label	Key Sub-Tabs / Views	Target User Roles
Authentication	`/login`	Secure Login	Show/Hide Password, Credential Authentication	All Roles
Operations	`/dashboard`	Executive Dashboard	Factory OEE, Production Output, Scrap %, Active Alarms	Management, Supervisor
	`/monitoring`	Production Monitor	Real-time Machine Status (Running 🟢, Idle ⚪, Breakdown 🔴, Setup 🟡, Maintenance 🔵), Machine Card Detail Links	Operator, Supervisor
	`/schedule`	Gantt Schedule	Drag-and-Drop Batch Timelines, Line Loadings	Production Manager
	`/operator`	Operator Station	Touch Workstation, Step Execution, Scrap Counter	Machine Operator
Materials & Recipes	`/material-management`	Material Management	1. Material Master 2. Qualified Vendors 3. Goods Receipt (GRN) 4. FEFO Stock & Location 5. Weighing & Dispensing Station	Warehouse, Supervisor, QA
	`/recipes`	Recipes & Formulas	Versioned Master Formulas, BOM Specification, GAMP 5 CPP Instructions, 3D STL/Image Preview	Formulator, Production Manager
Quality & Compliance	`/work-orders`	Batch Records	MBR Initiation, Campaign Run States (PLANNED ➔ RELEASED ➔ IN_PROGRESS ➔ COMPLETED ➔ CLOSED), EBR PDF Export	Production Manager, QA
	`/quality`	Quality & SPC	In-process QC Inspections, Deviation Management, CAPA Tracker, SPC X-bar/R Charts	Quality Assurance, QA Engineer
	`/equipment`	Equipment & Calibrations	1. Asset Register 2. NIST Calibration Certs 3. IQ/OQ/PQ Validation Protocols 4. Daily Calibration Checklists	Maintenance, Validation Lead
	`/compliance`	Compliance & Audits	21 CFR Part 11 Audit Trail, Cryptographic Verification, Electronic Signature History	Compliance Auditor, QA Lead
Track & Trace	`/trace/dashboard`	Trace Dashboard	FSMA 204 KPIs, Active Recall Risk, Serialization Stats	Supply Chain, QA
	`/trace/products`	Product Registry	GS1 GTIN Catalog, Storage Specs	Supply Chain, Master Data
	`/trace/batches`	Batch Campaigns	Work Order Lot Tracing, Expiry Tracking	Supply Chain, Production
	`/trace/trace`	Genealogy Trace	Forward & Backward Trace Tree, Interactive Flow	QA, Compliance Officer
	`/trace/events`	Milestone Events	Critical Tracking Events (CTE) &

Key Data Elements (KDE) Supply Chain, Auditor
`/trace/shipments` Shipments Ledger Outbound Dispatch, SSCC-18 Pallet Serialization Warehouse Logistics
`/trace/recall` Recall Simulator Targeted Batch Recall Simulator & Impact Assessment QA Director, Legal
Administration `/admin/material` Material Master Create & Register Materials System Admin
`/admin/equipment` Equipment Assets Plant Hierarchy, Area & Machine Allocation System Admin
`/admin/types-uom` Material Types & UOM Dynamic Material Categories & Measurement Units Catalog System Admin
`/admin/users` Users & Roles Account Roster, GAMP 5 Role Matrix, Password Resets System Admin
`/admin/standards` Standards & Serialization GS1, 21 CFR Part 11, EPCIS Webhook Configuration System Admin
`/admin/api-docs` API Integration Guide Interactive REST API Sandbox Playground Developers, Integrators
Support `/help` Help Center Standard Operating Procedures (SOPs), MES User Guide All Roles

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4. Global Navigation Shell & Header Layout

```
+-------------------------------------------------------------------------------------------------------+
| [LOGO] LUMOS MES - Next Gen Manufacturing Execution System     [TOUCH MODE] [DARK/LIGHT]  [🔔 (ALERTS)] [USER PROFILE | LOGOUT] |
+-----------------------------------+-------------------------------------------------------------------+
|  LEFT ACCORDION SIDEBAR           |  MAIN WORKSPACE VIEWPORT                                          |
|                                   |                                                                   |
|  ------------------------         |  ------------------------                                         |
|                                   |                                                                   |
|  ▼ Operations                     |  Breadcrumbs / Page Title                                         |
|                                   |                                                                   |
|    • Executive Dashboard          |                                                                   |
|                                   |                                                                   |
|    • Production Monitor           |                                                                   |
+-----------------------------------+-------------------------------------------------------------------+  |
|    • Gantt Schedule               |  | Content Cards / Interactive Tables / Modals                    |  |
|    • Operator Station             |  |                                                                |  |
|                                   |  |                                                                |  |
|  ► Materials & Recipes            |  |                                                                |  |
|                                   |  |                                                                |  |
|  ► Quality & Compliance           |  |                                                                |  |
|                                   |  |                                                                |  |
|  ► Track & Trace                  |  |                                                                |  |
|                                   |  |                                                                |  |
|  ► Administration                 |  |                                                                |  |
|                                   |  |                                                                |  |
```

```
|  ► Support                          |  |
                                    |  |
|                                     |
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```
Global Header Controls:
Touch / Factory Mode Switcher: Switches the UI between the high-density desktop management layout and high-contrast, large touch-target display for ruggedized shopfloor tablets.
Light / Dark Mode Switcher: Toggles dynamic theme styling with instantaneous CSS class and `data-theme` updates.
Interactive Alarms & Notifications Drawer:
Displays real-time WebSocket alarms with severity indicators (Critical 🔴, Warning 🟡, Info 🔵).
Click-Outside Dismissal: Clicking anywhere on the screen closes the drawer.
Click-Through Navigation: Clicking an alarm automatically marks it as read, closes the popover, and navigates directly to the affected equipment or process view.
Badge Clearing: The red unread counter badge automatically decrements and hides when all alarms are read.
Active User Profile & Logout: Displays current user name, GAMP 5 role, and secure session teardown.
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5. Key Cross-Page Navigation Flows
Flow A: Equipment Monitoring ➔ Asset Calibration View
Operator navigates to Production Monitor (`/monitoring`).
Identifies a machine flagged with 🟡 Maintenance / Calibration status (e.g., `EQ002` Steam Kettle).
Clicks the equipment card or the `"View Calibrations ➔"` link.
The system navigates to `/equipment?code=EQ002`.
The Equipment & Calibrations page highlights `EQ002`, displays a focused asset header banner, and auto-scrolls directly to the machine's calibration records.
```mermaid
sequenceDiagram
    autonumber
    actor User as Operator / Supervisor
    participant Mon as /monitoring (Production Monitor)
    participant Eq as /equipment (Equipment & Calibrations)

    User->>Mon: Inspects machine status cards
    User->>Mon: Clicks machine card (e.g. EQ002)
    Mon->>Eq: Navigates to /equipment?code=EQ002
    Eq->>Eq: Reads search param (?code=EQ002)
    Eq->>Eq: Focuses EQ002 row with glowing border & auto-scrolls
    Eq-->>User: Displays focused NIST calibration & IQ/OQ/PQ protocols
```

Flow B: Batch Campaign Execution ➔ Dispensing ➔ EBR Release
Production Manager initiates a new Batch Campaign in Batch Records

(`/work-orders`).
Transitions work order status: `PLANNED` ➔ `RELEASED` with a 21 CFR Part 11 Electronic Signature.
Warehouse operator navigates to Material Management (`/material-management` ➔ Weighing & Dispensing Station).
Selects the active Batch Campaign and triggers FEFO (First-Expired, First-Out) stock allocation.
Performs weighing balance with dual-operator electronic signoff.
Once production finishes, supervisor transitions batch to `COMPLETED` and QA signs off `CLOSED` to generate the Electronic Batch Record (EBR) PDF dossier.

```mermaid
sequenceDiagram
    autonumber
    actor Mgr as Production Manager
    actor Op as Operator / QA
    participant WO as /work-orders (Batch Records)
    participant Disp as /material-management (Dispensing)
    participant Sign as /compliance/sign (21 CFR Part 11)

    Mgr->>WO: Creates & Releases MBR Batch
    WO->>Sign: Authorizes status transition to RELEASED
    Op->>Disp: Selects Batch & executes FEFO stock allocation
    Disp->>Sign: Performs double-signature weighing verification
    Op->>WO: Completes production run (Status: COMPLETED)
    Mgr->>Sign: QA Approval & Dossier Seal (Status: CLOSED)
    WO-->>Mgr: Downloads verified EBR Dossier PDF
```

Flow C: Material Master & Dynamic Catalog Maintenance
Administrator opens Administration ➔ Material Types & UOM (`/admin/types-uom`).
Adds a new category (e.g., `EXCIPIENT`) or Unit of Measure (e.g., `ML`).
The catalog immediately persists into the database and syncs across:
Admin Material Master (`/admin/material`)
Material Master Definition Modal (`/material-management`)
Recipe Bill of Materials (BOM) (`/recipes`)

```mermaid
sequenceDiagram
    autonumber
    actor Admin as System Administrator
    participant Types as /admin/types-uom (Catalog Config)
    participant MatMgmt as /material-management (Material Management)
    participant Rec as /recipes (Recipes & Formulas)

    Admin->>Types: Registers new Material Type / UOM
    Types->>Types: Persists to /materials/types and /materials/uoms
    Admin->>MatMgmt: Opens Add Material modal
    MatMgmt->>MatMgmt: Dynamically loads updated dropdown options
    Admin->>Rec: Opens Recipe Designer
    Rec->>Rec: BOM dropdowns reflect updated material definitions
```

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6. Security & Authorization Matrix

User Role	Operations	Materials & Recipes	Quality & Compliance	Track & Trace	Administration
ADMIN	Full View	Full CRUD	Full CRUD + Sign	Full CRUD	Full Access
PRODUCTION_MANAGER	Full Control	Full CRUD	Release & Close MBR	Full Control	Read-Only
SUPERVISOR	Line Execution	View & Dispense Batch Start / Pause		Full View	Read-Only
OPERATOR	Touch Console	Dispense Run	Read / Log Scrap	View Genealogy	No Access
QUALITY_ENGINEER	View	GRN Inspection Release	QA Signoff / CAPA / EBR	Recall Simulator	No Access